JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI SAKTHI CORPORATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

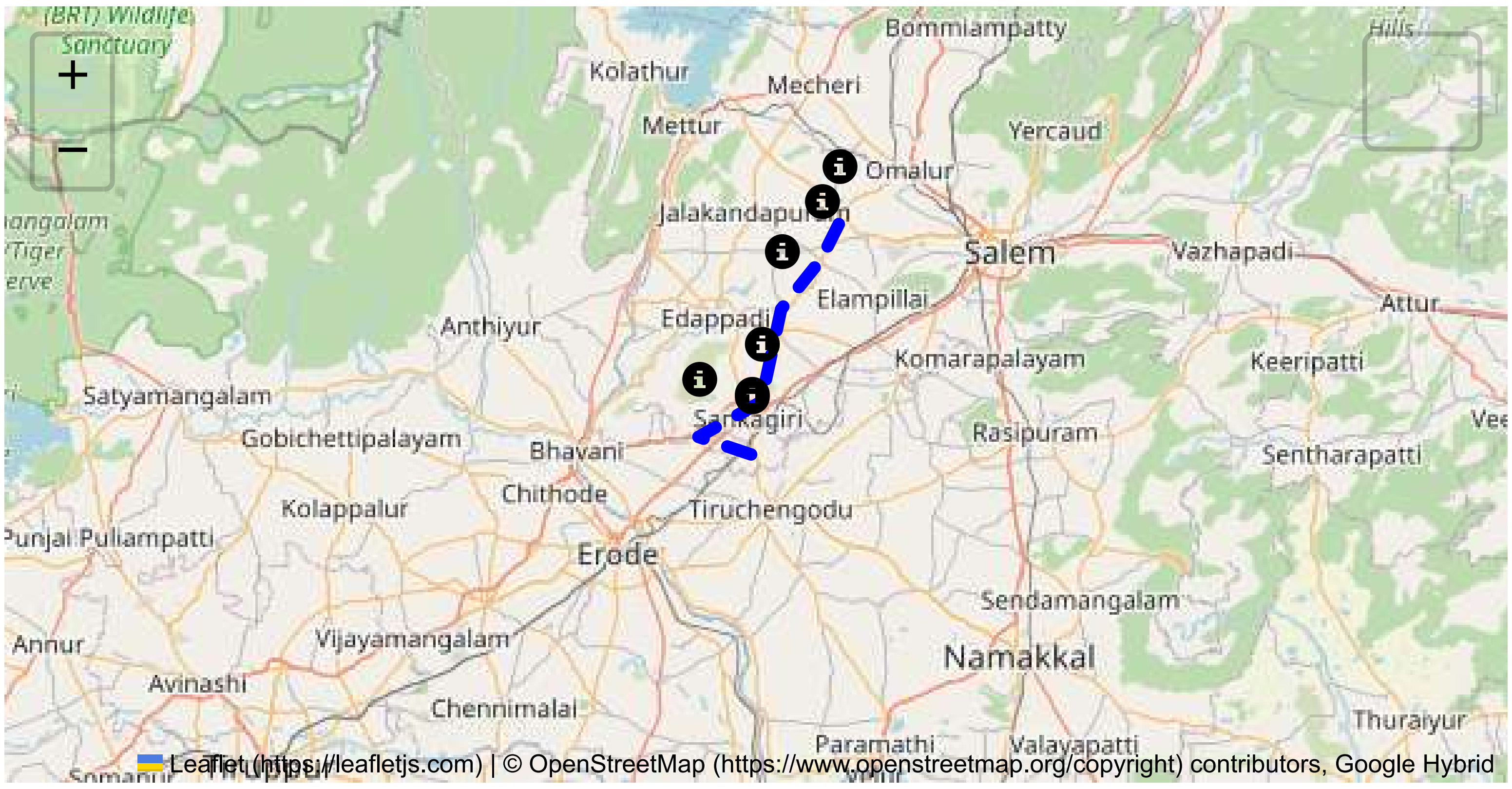
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

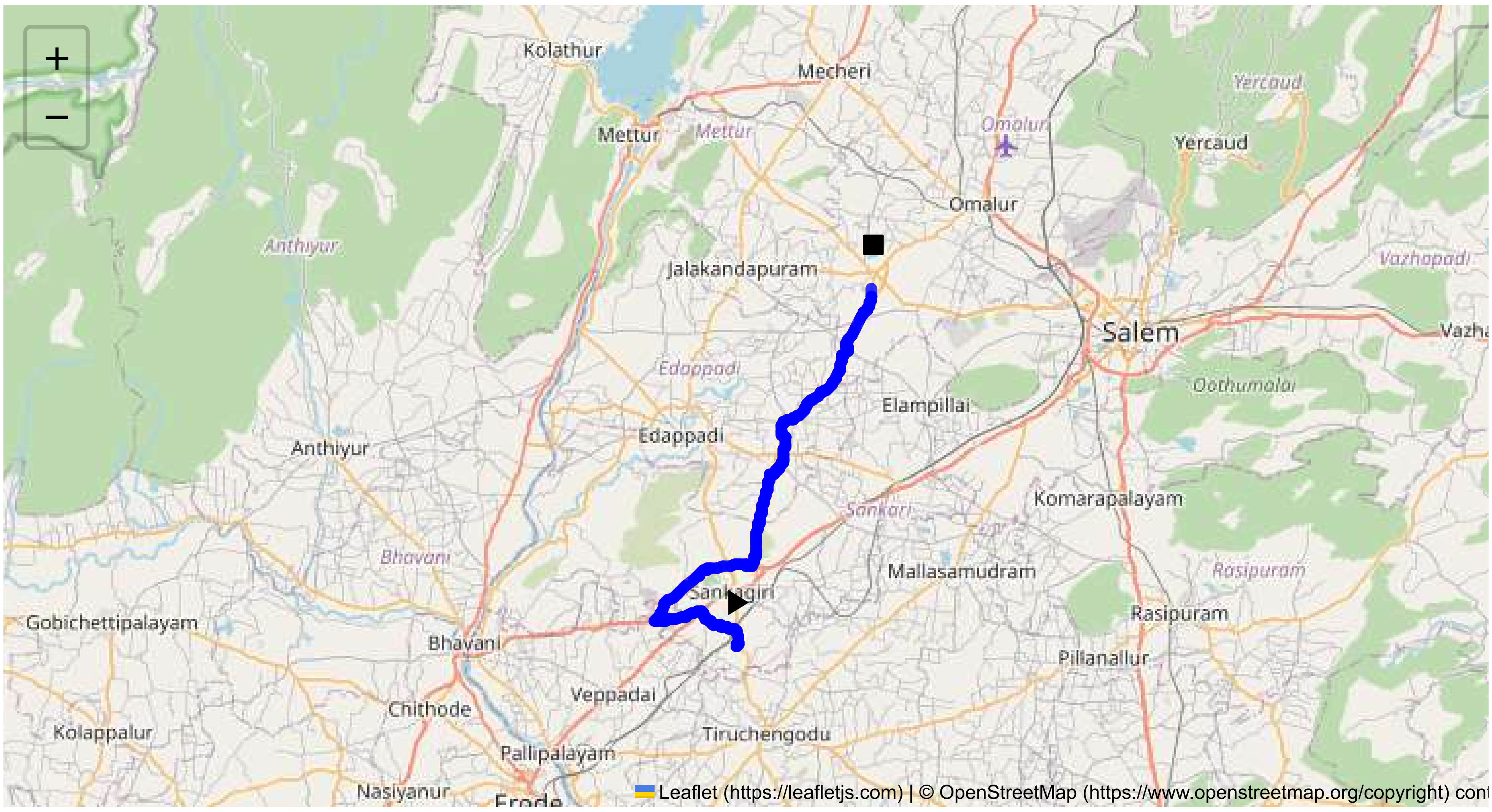
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.60 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.684896, 77.969032)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 42.60 kilometers across Tamil Nadu, starting from Sangagiri and passing through Puthur, Padaiveedu, Edappadi, and Pappampadi before reaching Omalur, Tharamangalam. The route utilizes predominantly state highways and local roads, making it feasible for truck travel.

2. Typical Weather Conditions and Potential Weather-related Hazards

Tamil Nadu experiences a tropical climate with a hot and dry phase typically from March to June, the monsoon season from July to November, and a cooler period from December to February. During the monsoon, the region may encounter heavy rain, leading to water-logged roads or increased risk of landslides. Monitoring weather updates during these periods is recommended.

3. Analysis of Traffic Patterns

Traffic along this route is moderate, with congestion likely occurring during mornings (7:30 AM to 9:30 AM) and evenings (4:30 PM to 7:30 PM) as local residents commute. Key congestion spots include Sangagiri town center and crossings near Edappadi due to school and market activities.

4. Assessment of Road Quality and Infrastructure

The state highways within this route are generally well-maintained. However, rural roads between Pappampadi and Tharamangalam may have varied quality with potential for potholes or narrow lanes. Awareness of construction or road repair signs is crucial, as these can cause sudden route diversions.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes include using NH-544 from Sangagiri toward Salem and then connecting to the Salem-Coimbatore Highway towards Tharamangalam if the primary route is blocked.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Local authorities require pertinent permits for transporting hazardous materials through populated areas. Trucks might be required to use designated routes avoiding city centers, so knowledge of these paths is crucial.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historically, the region has seen occasional incidents of heavy vehicle accidents, mainly due to speeding or vehicular failures. Rigorous maintenance checks for vehicles and adherence to speed limits have significantly mitigated risks.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural regions, demanding careful handling of hazardous materials to prevent contamination of water bodies and farmland. Proximity to the Kaveri River basin also necessitates caution to prevent spills during transport.

9. Analysis of Communication Coverage

Generally, mobile network coverage is adequate throughout the route. However, minor dead zones may exist in some rural segments near Pappampadi, which could impact communication, requiring prior communication before entering these zones.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response times in urban areas and along major highways are approximately 15-30 minutes. However, this may extend to up to 45 minutes in rural segments with limited accessibility.

11. Overall Summary of Risk Assessment

The journey from Sangagiri to Omalur via the specified route presents a moderate risk level due to variable road conditions and weather patterns. Drivers should remain vigilant of traffic during peak hours, particularly in urban and rural town centers. Strict adherence to transport regulations, systematic route planning with alternative paths, and contingency measures for potential hazards will enhance safety during transport. Regular updates from weather services and local traffic departments should be observed to preempt any sudden changes affecting journey safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43968, 77.87345	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.49409, 77.88004	15 KM/Hr
12	Turn	Medium	11.49419, 77.88015	30 KM/Hr
13	Turn	High	11.49383, 77.88442	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.49419, 77.88454	30 KM/Hr
15	Turn	High	11.49448, 77.88490	15 KM/Hr
16	Turn	High	11.55614, 77.89764	15 KM/Hr
17	Turn	High	11.58320, 77.90646	15 KM/Hr
18	Turn	Medium	11.64197, 77.95197	30 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88454

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49448, 77.88490

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55614, 77.89764

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58320, 77.90646

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64197, 77.95197